

Global convergence theory for Anderson mixing methods for Euclidean fixed-point problems

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Website #4 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

We are given a map $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and the fixed-point problem

$$g(x^*) = x^*.$$

Write the residual as

$$r(x) := g(x) - x.$$

For a fixed memory depth m , Anderson mixing (type-II / classical Anderson acceleration) uses

$$\Delta x_j := x_{j+1} - x_j, \quad \Delta r_j := r(x_{j+1}) - r(x_j),$$

and, at iteration k , with the standard startup convention $m_k := \min\{m, k\}$,

$$X_k = [\Delta x_{k-m_k}, \dots, \Delta x_{k-1}] \in \mathbb{R}^{n \times m_k}, \quad R_k = [\Delta r_{k-m_k}, \dots, \Delta r_{k-1}] \in \mathbb{R}^{n \times m_k}.$$

Choose any least-squares minimizer

$$\Gamma_k \in \arg \min_{\Gamma \in \mathbb{R}^{m_k}} \|r(x_k) - R_k \Gamma\|_2,$$

and set

$$\bar{x}_k := x_k - X_k \Gamma_k, \quad \bar{r}_k := r(x_k) - R_k \Gamma_k, \quad x_{k+1} := \bar{x}_k + \beta_k \bar{r}_k,$$

with $\beta_k > 0$.

The problem asks for verifiable assumptions on g and on the parameters $(m, \{\beta_k\})$ under which, for arbitrary $x_0 \in \mathbb{R}^n$, the Anderson sequence is guaranteed to converge to a fixed point, or at least to have a subsequence converging to a fixed point.

Alternative readings considered

- The displayed matrices X_k, R_k in the prompt only make literal sense once $k \geq m$. I adopt the standard startup convention $m_k = \min\{m, k\}$, which is the strongest natural reading that keeps the algorithm well-defined from $k = 0$.
- The least-squares problem may have multiple minimizers. I therefore formulate all nonlinear results so that the hypotheses are imposed on the *actual* minimizer selected by the algorithm at each step.
- I treat the strongest interpretation first: the exact unsafeguarded type-II scheme above. When I cannot prove whole-sequence convergence, I explicitly downgrade to subsequential convergence and say so.

Related literature found

I found no source claiming a broad global convergence theorem for the *exact*, unsafeguarded, limited-memory type-II Anderson scheme over a general nonlinear class. The sources I found split into affine/global-linear results, local nonlinear results with coefficient bounds, and safeguarded or constrained variants.

- **Saad (2025)** — BACKGROUND. A recent survey that cleanly maps the terrain: affine/global linear theory, local nonlinear theory under bounded coefficients, and safeguarded variants. It was especially useful for locating the exact frontier of what is known.
- **Walker–Ni (2011)** — TOOL. Shows full-memory linear AA is essentially equivalent to GMRES and type-I to Arnoldi. Very relevant for orientation in the affine case, but not itself a global theorem for the exact limited-memory nonlinear scheme of the prompt.
- **Toth–Kelley (2015)** — PARTIAL. Gives local r -linear convergence for contractive nonlinear maps under bounded combination coefficients, and (as summarized in the 2025 review) a global q -linear residual decrease for affine contractive maps. This is the closest pre-existing result to my part (A) and a conceptual precursor of my part (B).
- **Zhang–O’Donoghue–Boyd (2020)** — PARTIAL. Proves global convergence for a *safeguarded type-I* Anderson variant under nonexpansiveness. Highly relevant conceptually, but it is a modified algorithm (safeguarding, Powell regularization, restarts) and not the exact type-II scheme here.
- **Barré–Taylor–d’Aspremont (2022)** — PARTIAL. Studies a constrained vector-extrapolation / constrained-AA scheme with explicit bounds when extrapolation weights are controlled. This directly suggested my coefficient-control route in the nonlinear contractive case, but their algorithm is constrained/stabilized rather than the raw least-squares scheme here.
- **Pollock–Rebholz (2021)** — TOOL. Gives one-step residual bounds and motivates safeguarding and depth control. Useful methodologically, but not a global theorem for the exact unsafeguarded scheme.
- **Bian–Chen–Kelley (2021)** — PARTIAL. Proves convergence for a special nonsmooth class with a smooth contractive part plus a small-Lipschitz nonsmooth part. Promising, but too structured for the general fixed-point problem posed here.

Proof sketch

There are three regimes.

First, for *affine* maps $g(x) = Mx + b$, the residual is affine:

$$r(x) = (M - I)x + b.$$

This makes the Anderson-corrected residual $\bar{r}_k$ equal to the *true residual at the corrected point* $\bar{x}_k$. That identity is special to the affine case and is the key simplification. Then the next residual is obtained by multiplying $\bar{r}_k$ by the relaxed linear map $(1 - \beta_k)I + \beta_k M$. Since $\Gamma = 0$ is always feasible in the least-squares problem, $\|\bar{r}_k\|_2 \leq \|r(x_k)\|_2$, and global q -linear residual decay follows immediately whenever the relaxed linear map is uniformly contractive.

Second, for a *nonlinear contraction*, the affine identity $\bar{r}_k = r(\bar{x}_k)$ is no longer true, so a residual proof breaks. The right move is to rewrite Anderson mixing in its equivalent α -form:

$$x_{k+1} = \sum_i \alpha_i^{(k)} T_{\beta_k}(x_i), \quad T_{\beta}(x) := (1 - \beta)x + \beta g(x).$$

The map T_{β} is a contraction with factor

$$q(\beta) = |1 - \beta| + \beta c$$

whenever g is c -contractive. Thus the only extra price paid by Anderson is the ℓ_1 -size of the combination coefficients:

$$\|x_{k+1} - x^*\|_2 \leq q(\beta_k) \|\alpha^{(k)}\|_1 \max_{\text{window}} \|x_j - x^*\|_2.$$

If the *effective Anderson gain*

$$\eta_k := q(\beta_k) \|\alpha^{(k)}\|_1$$

is uniformly < 1 , then a block-maximum argument over blocks of length $m + 1$ gives global R -linear convergence. This is fully rigorous and requires no differentiability.

Third, for a *nonexpansive* map, global whole-sequence convergence of the exact memory method is harder. But if the Anderson coefficients are convex ($\alpha_i^{(k)} \geq 0$, $\sum_i \alpha_i^{(k)} = 1$) and the current iterate keeps a positive weight, then the step is a convex combination of averaged-operator images $T_{\beta_k}(x_i)$. A direct Euclidean identity yields

$$\|T_{\beta}(x) - z\|_2^2 \leq \|x - z\|_2^2 - \beta(1 - \beta) \|r(x)\|_2^2$$

for any fixed point z . Combining this with convexity of $\|\cdot\|_2^2$ gives blockwise boundedness and forces arbitrarily small residuals to occur infinitely often; compactness then yields a subsequence converging to a fixed point.

The actionable parameter recommendations coming out of the proof are:

- in the contractive regimes, $\beta_k \equiv 1$ is the best constant damping in this theory, since $q(\beta)$ is minimized at $\beta = 1$ and equals c ;
- in the nonexpansive regime, keep β_k in a compact subinterval of $(0, 1)$, e.g. $\beta_k \equiv \frac{1}{2}$;
- the memory m may be arbitrary and finite; what must be controlled is not m itself but the coefficient behavior it induces.

Main result

Theorem 1. *For each k , let $m_k := \min\{m, k\}$, write*

$$\Gamma_k = (\gamma_0^{(k)}, \dots, \gamma_{m_k-1}^{(k)})^\top \in \mathbb{R}^{m_k},$$

and define coefficients $\alpha^{(k)} = (\alpha_0^{(k)}, \dots, \alpha_{m_k}^{(k)})$ by

$$\alpha_0^{(k)} = \gamma_0^{(k)}, \quad \alpha_i^{(k)} = \gamma_i^{(k)} - \gamma_{i-1}^{(k)} \quad (1 \leq i \leq m_k - 1), \quad \alpha_{m_k}^{(k)} = 1 - \gamma_{m_k-1}^{(k)}$$

when $m_k \geq 1$, and $\alpha_0^{(k)} = 1$ when $m_k = 0$.

Then the Anderson step admits the equivalent representation

$$x_{k+1} = \sum_{i=0}^{m_k} \alpha_i^{(k)} T_{\beta_k}(x_{k-m_k+i}), \quad T_{\beta}(x) := (1 - \beta)x + \beta g(x),$$

and $\sum_{i=0}^{m_k} \alpha_i^{(k)} = 1$.

Under this notation, the following hold.

(A) Affine global convergence. Assume $g(x) = Mx + b$ is affine and

$$\rho := \sup_{k \geq 0} \|(1 - \beta_k)I + \beta_k M\|_2 < 1.$$

Then for every initial point $x_0 \in \mathbb{R}^n$, for every memory depth m , and for every choice of least-squares minimizers Γ_k , the Anderson iterates are well-defined and converge to the unique fixed point

$$x^* = (I - M)^{-1}b.$$

Moreover,

$$\|r(x_k)\|_2 \leq \rho^k \|r(x_0)\|_2 \quad \text{for all } k \geq 0,$$

and therefore

$$\|x_k - x^*\|_2 \leq \|(I - M)^{-1}\|_2 \rho^k \|r(x_0)\|_2.$$

(B) Nonlinear contractive global convergence under coefficient control. Assume g is globally c -contractive in $\|\cdot\|_2$, with $0 \leq c < 1$, and define

$$q_k := |1 - \beta_k| + \beta_k c.$$

If

$$\eta := \sup_{k \geq 0} q_k \|\alpha^{(k)}\|_1 < 1,$$

then g has a unique fixed point x^* , and the Anderson iterates converge globally R -linearly to x^* . More precisely,

$$\|x_k - x^*\|_2 \leq \eta^{\lfloor k/(m+1) \rfloor} \max_{0 \leq j \leq m} \|x_j - x^*\|_2 \quad (k \geq 0).$$

In addition,

$$(1 - c)\|x_k - x^*\|_2 \leq \|r(x_k)\|_2 \leq (1 + c)\|x_k - x^*\|_2,$$

so the residuals converge R -linearly as well.

A simpler sufficient condition is: there exists $M \geq 1$ such that $\|\alpha^{(k)}\|_1 \leq M$ for all k , and

$$\sup_{k \geq 0} q_k < \frac{1}{M}.$$

In particular, if the coefficients are convex, i.e. $\alpha_i^{(k)} \geq 0$ and $\sum_i \alpha_i^{(k)} = 1$, then $\|\alpha^{(k)}\|_1 = 1$ and global convergence follows whenever $\sup_k q_k < 1$ (for instance, under the constant choice $\beta_k \equiv 1$, where $q_k = c$).

(C) *Nonexpansive global subsequential convergence under convex-combination control.*

Assume g is nonexpansive in $\|\cdot\|_2$, the fixed-point set $\{x : g(x) = x\}$ is nonempty, and

$$\beta_k \in [\underline{\beta}, \overline{\beta}] \subset (0, 1) \quad \text{for all } k.$$

Assume also that for every k ,

$$\alpha_i^{(k)} \geq 0, \quad \sum_{i=0}^{m_k} \alpha_i^{(k)} = 1, \quad \alpha_{m_k}^{(k)} \geq \underline{\alpha} > 0.$$

Then the Anderson iterates are bounded, and for every $\varepsilon > 0$ there are infinitely many indices k such that

$$\|r(x_k)\|_2 < \varepsilon.$$

Consequently, the sequence $\{x_k\}$ has a subsequence converging to a fixed point of g .

Proof

From now on, $\|\cdot\|$ means $\|\cdot\|_2$.

Lemma 1. *For every k , the Γ_k -form and the $\alpha^{(k)}$ -form are equivalent:*

$$\bar{x}_k = \sum_{i=0}^{m_k} \alpha_i^{(k)} x_{k-m_k+i}, \quad \bar{r}_k = \sum_{i=0}^{m_k} \alpha_i^{(k)} r(x_{k-m_k+i}),$$

and therefore

$$x_{k+1} = \sum_{i=0}^{m_k} \alpha_i^{(k)} \left(x_{k-m_k+i} + \beta_k r(x_{k-m_k+i}) \right) = \sum_{i=0}^{m_k} \alpha_i^{(k)} T_{\beta_k}(x_{k-m_k+i}).$$

Also,

$$\sum_{i=0}^{m_k} \alpha_i^{(k)} = 1.$$

Proof. If $m_k = 0$, then there is no least-squares correction, $\bar{x}_k = x_k$, $\bar{r}_k = r(x_k)$, and the claims are immediate.

Assume $m_k \geq 1$. By definition,

$$\bar{x}_k = x_k - \sum_{j=0}^{m_k-1} \gamma_j^{(k)} \Delta x_{k-m_k+j} = x_k - \sum_{j=0}^{m_k-1} \gamma_j^{(k)} (x_{k-m_k+j+1} - x_{k-m_k+j}).$$

Collecting the coefficients of $x_{k-m_k}, x_{k-m_k+1}, \dots, x_k$, we obtain

$$\bar{x}_k = \gamma_0^{(k)} x_{k-m_k} + \sum_{i=1}^{m_k-1} (\gamma_i^{(k)} - \gamma_{i-1}^{(k)}) x_{k-m_k+i} + (1 - \gamma_{m_k-1}^{(k)}) x_k,$$

which is exactly

$$\bar{x}_k = \sum_{i=0}^{m_k} \alpha_i^{(k)} x_{k-m_k+i}.$$

The same telescoping calculation, with $r(x_j)$ in place of x_j , gives

$$\bar{r}_k = r(x_k) - \sum_{j=0}^{m_k-1} \gamma_j^{(k)} \Delta r_{k-m_k+j} = \sum_{i=0}^{m_k} \alpha_i^{(k)} r(x_{k-m_k+i}).$$

Finally,

$$x_{k+1} = \bar{x}_k + \beta_k \bar{r}_k = \sum_{i=0}^{m_k} \alpha_i^{(k)} \left(x_{k-m_k+i} + \beta_k r(x_{k-m_k+i}) \right),$$

which is the claimed T_{β_k} -form. The identity $\sum_i \alpha_i^{(k)} = 1$ is immediate from the telescoping definition:

$$\sum_{i=0}^{m_k} \alpha_i^{(k)} = \gamma_0^{(k)} + \sum_{i=1}^{m_k-1} (\gamma_i^{(k)} - \gamma_{i-1}^{(k)}) + 1 - \gamma_{m_k-1}^{(k)} = 1.$$

□

Lemma 2. Assume g is globally c -contractive, $0 \leq c < 1$:

$$\|g(u) - g(v)\| \leq c\|u - v\| \quad \text{for all } u, v \in \mathbb{R}^n.$$

Then:

(i) g has a unique fixed point $x^* \in \mathbb{R}^n$;

(ii) for every $\beta > 0$, the relaxed map

$$T_\beta(x) := (1 - \beta)x + \beta g(x)$$

is Lipschitz with constant

$$q(\beta) := |1 - \beta| + \beta c;$$

(iii) for every $x \in \mathbb{R}^n$,

$$(1 - c)\|x - x^*\| \leq \|r(x)\| \leq (1 + c)\|x - x^*\|.$$

Proof. For (i), start Picard iteration from any $y_0 \in \mathbb{R}^n$:

$$y_{t+1} := g(y_t).$$

Then

$$\|y_{t+1} - y_t\| = \|g(y_t) - g(y_{t-1})\| \leq c\|y_t - y_{t-1}\|.$$

Inductively,

$$\|y_{t+1} - y_t\| \leq c^t \|y_1 - y_0\|.$$

Hence, for $s > t$,

$$\|y_s - y_t\| \leq \sum_{j=t}^{s-1} \|y_{j+1} - y_j\| \leq \sum_{j=t}^{s-1} c^j \|y_1 - y_0\| \leq \frac{c^t}{1 - c} \|y_1 - y_0\|.$$

Therefore $\{y_t\}$ is Cauchy in $\mathbb{R}^n$, hence convergent to some x^* . Since g is Lipschitz, it is continuous, so

$$g(x^*) = g\left(\lim_{t \rightarrow \infty} y_t\right) = \lim_{t \rightarrow \infty} g(y_t) = \lim_{t \rightarrow \infty} y_{t+1} = x^*.$$

Thus $x^\star$ is a fixed point. If $z^\star$ is another fixed point, then

$$\|x^\star - z^\star\| = \|g(x^\star) - g(z^\star)\| \leq c\|x^\star - z^\star\|.$$

Since $c < 1$, this implies $\|x^\star - z^\star\| = 0$, so $x^\star = z^\star$.

For (ii), for arbitrary u, v ,

$$\begin{aligned} \|T_\beta(u) - T_\beta(v)\| &= \|(1 - \beta)(u - v) + \beta(g(u) - g(v))\| \\ &\leq |1 - \beta| \|u - v\| + \beta\|g(u) - g(v)\| \leq (|1 - \beta| + \beta c)\|u - v\|. \end{aligned}$$

For (iii), since $r(x^\star) = 0$,

$$r(x) = g(x) - x = (g(x) - g(x^\star)) - (x - x^\star).$$

Therefore

$$\|r(x)\| \leq \|g(x) - g(x^\star)\| + \|x - x^\star\| \leq (1 + c)\|x - x^\star\|.$$

For the lower bound,

$$\|r(x)\| = \|(x - x^\star) - (g(x) - g(x^\star))\| \geq \|x - x^\star\| - \|g(x) - g(x^\star)\| \geq (1 - c)\|x - x^\star\|.$$

□

Lemma 3. *Let $\{e_k\}_{k \geq 0} \subset [0, \infty)$ satisfy*

$$e_{k+1} \leq \eta \max_{k-m_k \leq j \leq k} e_j \quad (k \geq 0),$$

where $m_k = \min\{m, k\}$ and $0 \leq \eta < 1$. Define block starts

$$N_\ell := \ell(m + 1), \quad \ell = 0, 1, 2, \dots,$$

and block maxima

$$E_\ell := \max_{N_\ell \leq j \leq N_\ell + m} e_j.$$

Then

$$E_{\ell+1} \leq \eta E_\ell \quad (\ell \geq 0).$$

Consequently,

$$e_k \leq \eta^{\lfloor k/(m+1) \rfloor} E_0 \quad (k \geq 0).$$

Proof. Fix $\ell \geq 0$. We prove by induction on $s = 1, \dots, m + 1$ that

$$e_{N_{\ell+1}+s-1} \leq \eta E_\ell.$$

For such an s , let

$$k := N_\ell + m + s - 1.$$

Then $k + 1 = N_{\ell+1} + s - 1$. Every index j appearing in the recurrence for e_{k+1} satisfies

$$k - m_k \leq j \leq k.$$

Since $m_k \leq m$, we have

$$k - m_k \geq k - m = N_\ell + s - 1,$$

and therefore

$$j \in [N_\ell + s - 1, N_\ell + m + s - 1].$$

This interval is contained in the union of

$$[N_\ell, N_\ell + m] \quad \text{and} \quad [N_{\ell+1}, N_{\ell+1} + s - 2].$$

Hence every such e_j is either in the previous block, where $e_j \leq E_\ell$, or among the earlier entries of the new block, which have already been shown (by the induction hypothesis) to satisfy $e_j \leq \eta E_\ell \leq E_\ell$. Thus

$$\max_{k-m_k \leq j \leq k} e_j \leq E_\ell.$$

Applying the assumed recurrence,

$$e_{N_{\ell+1}+s-1} = e_{k+1} \leq \eta E_\ell.$$

This proves the induction. Taking the maximum over $s = 1, \dots, m + 1$ yields

$$E_{\ell+1} \leq \eta E_\ell.$$

Iterating,

$$E_\ell \leq \eta^\ell E_0.$$

Finally, if k belongs to block $\ell = \lfloor k/(m+1) \rfloor$, then $e_k \leq E_\ell$, hence

$$e_k \leq \eta^{\lfloor k/(m+1) \rfloor} E_0.$$

□

Lemma 4. *Assume g is nonexpansive:*

$$\|g(u) - g(v)\| \leq \|u - v\| \quad \text{for all } u, v \in \mathbb{R}^n.$$

Let z be a fixed point of g , and let $\beta \in (0, 1)$. Then

$$\|T_\beta(x) - z\|^2 \leq \|x - z\|^2 - \beta(1 - \beta)\|r(x)\|^2 \quad \text{for all } x \in \mathbb{R}^n.$$

Proof. Since $g(z) = z$,

$$T_\beta(x) - z = (1 - \beta)(x - z) + \beta(g(x) - z).$$

For arbitrary vectors a, b and $\beta \in (0, 1)$,

$$\|(1 - \beta)a + \beta b\|^2 = (1 - \beta)\|a\|^2 + \beta\|b\|^2 - \beta(1 - \beta)\|a - b\|^2.$$

Apply this identity with

$$a := x - z, \quad b := g(x) - z.$$

Then

$$a - b = x - g(x) = -r(x),$$

hence

$$\|T_\beta(x) - z\|^2 = (1 - \beta)\|x - z\|^2 + \beta\|g(x) - z\|^2 - \beta(1 - \beta)\|r(x)\|^2.$$

Because g is nonexpansive and $g(z) = z$,

$$\|g(x) - z\| = \|g(x) - g(z)\| \leq \|x - z\|.$$

Therefore

$$\|T_\beta(x) - z\|^2 \leq \|x - z\|^2 - \beta(1 - \beta)\|r(x)\|^2.$$

□

Proof of the theorem. We prove parts **(A)**, **(B)**, and **(C)** in turn.

Part (A): affine maps.

Assume $g(x) = Mx + b$. Then

$$r(x) = g(x) - x = (M - I)x + b.$$

Set

$$A := M - I.$$

Since

$$\Delta r_j = r(x_{j+1}) - r(x_j) = A(x_{j+1} - x_j) = A\Delta x_j,$$

we have

$$R_k = AX_k \quad \text{for every } k.$$

Hence

$$\bar{r}_k = r(x_k) - R_k \Gamma_k = Ax_k + b - AX_k \Gamma_k = A(x_k - X_k \Gamma_k) + b = A\bar{x}_k + b = r(\bar{x}_k).$$

This is the affine identity mentioned in the proof sketch.

Now compute the next residual:

$$r(x_{k+1}) = r(\bar{x}_k + \beta_k \bar{r}_k) = A(\bar{x}_k + \beta_k \bar{r}_k) + b = (A\bar{x}_k + b) + \beta_k A\bar{r}_k = \bar{r}_k + \beta_k A\bar{r}_k.$$

Since $A = M - I$,

$$I + \beta_k A = I + \beta_k (M - I) = (1 - \beta_k)I + \beta_k M.$$

Therefore

$$r(x_{k+1}) = ((1 - \beta_k)I + \beta_k M)\bar{r}_k.$$

Because $\Gamma = 0$ is always feasible in the least-squares problem,

$$\|\bar{r}_k\| = \min_{\Gamma} \|r(x_k) - R_k \Gamma\| \leq \|r(x_k) - R_k \cdot 0\| = \|r(x_k)\|.$$

Hence, by the definition of ρ ,

$$\|r(x_{k+1})\| \leq \|(1 - \beta_k)I + \beta_k M\| \cdot \|\bar{r}_k\| \leq \rho \|r(x_k)\|.$$

Inductively,

$$\|r(x_k)\| \leq \rho^k \|r(x_0)\|.$$

It remains to identify the fixed point and transfer residual convergence to iterate convergence. If $I - M$ were singular, there would exist $v \neq 0$ such that $Mv = v$. Then for every k ,

$$((1 - \beta_k)I + \beta_k M)v = v,$$

which implies

$$\|(1 - \beta_k)I + \beta_k M\| \geq 1,$$

contradicting $\rho < 1$. Thus $I - M$ is invertible, so

$$x^* = (I - M)^{-1}b$$

is the unique fixed point of g . Since

$$r(x) = (M - I)(x - x^*) = -(I - M)(x - x^*),$$

we have

$$x - x^* = -(I - M)^{-1}r(x),$$

and therefore

$$\|x_k - x^*\| \leq \|(I - M)^{-1}\| \cdot \|r(x_k)\| \leq \|(I - M)^{-1}\| \rho^k \|r(x_0)\|.$$

This proves part **(A)**.

Part (B): nonlinear contractions with coefficient control.

Assume g is globally c -contractive. By Lemma 2, g has a unique fixed point x^* , and each relaxed map T_{β_k} is q_k -Lipschitz with

$$q_k = |1 - \beta_k| + \beta_k c.$$

By Lemma 1,

$$x_{k+1} = \sum_{i=0}^{m_k} \alpha_i^{(k)} T_{\beta_k}(x_{k-m_k+i}).$$

Since x^* is also a fixed point of every T_{β_k} ,

$$x_{k+1} - x^* = \sum_{i=0}^{m_k} \alpha_i^{(k)} (T_{\beta_k}(x_{k-m_k+i}) - T_{\beta_k}(x^*)).$$

Therefore,

$$\begin{aligned} \|x_{k+1} - x^*\| &\leq \sum_{i=0}^{m_k} |\alpha_i^{(k)}| \|T_{\beta_k}(x_{k-m_k+i}) - T_{\beta_k}(x^*)\| \\ &\leq q_k \sum_{i=0}^{m_k} |\alpha_i^{(k)}| \max_{0 \leq i \leq m_k} \|x_{k-m_k+i} - x^*\| \leq \eta \max_{k-m_k \leq j \leq k} \|x_j - x^*\|. \end{aligned}$$

Set

$$e_k := \|x_k - x^*\|.$$

Then $e_{k+1} \leq \eta \max_{k-m_k \leq j \leq k} e_j$. Since $\eta < 1$, Lemma 3 gives

$$e_k \leq \eta^{\lfloor k/(m+1) \rfloor} \max_{0 \leq j \leq m} e_j,$$

which is the claimed global R -linear convergence.

The residual estimate follows from Lemma 2(iii):

$$(1 - c)\|x_k - x^*\| \leq \|r(x_k)\| \leq (1 + c)\|x_k - x^*\|.$$

Thus $\|r(x_k)\| \rightarrow 0$ R -linearly as well.

If, more simply, $\|\alpha^{(k)}\|_1 \leq M$ for every k , then

$$\eta \leq M \sup_k q_k,$$

so the hypothesis $\sup_k q_k < 1/M$ implies $\eta < 1$. Finally, if the coefficients are convex, then $\|\alpha^{(k)}\|_1 = 1$, and the condition reduces to $\sup_k q_k < 1$. This proves part **(B)**.

Part (C): nonexpansive maps with convex-combination control.

Assume g is nonexpansive and $\text{Fix}(g) \neq \emptyset$. Fix any $z \in \text{Fix}(g)$. Define

$$d_k := \|x_k - z\|^2, \quad \delta := \underline{\beta}(1 - \overline{\beta}) > 0.$$

By Lemma 1 and convexity of the function $u \mapsto \|u\|^2$,

$$d_{k+1} = \left\| \sum_{i=0}^{m_k} \alpha_i^{(k)} T_{\beta_k}(x_{k-m_k+i}) - z \right\|^2 \leq \sum_{i=0}^{m_k} \alpha_i^{(k)} \|T_{\beta_k}(x_{k-m_k+i}) - z\|^2.$$

Applying Lemma 4 termwise,

$$d_{k+1} \leq \sum_{i=0}^{m_k} \alpha_i^{(k)} \left(\|x_{k-m_k+i} - z\|^2 - \beta_k(1 - \beta_k) \|r(x_{k-m_k+i})\|^2 \right).$$

Since $\beta_k(1 - \beta_k) \geq \delta$,

$$d_{k+1} \leq \sum_{i=0}^{m_k} \alpha_i^{(k)} d_{k-m_k+i} - \delta \sum_{i=0}^{m_k} \alpha_i^{(k)} \|r(x_{k-m_k+i})\|^2. \quad (1)$$

Now define block starts

$$N_\ell := \ell(m+1)$$

and block maxima

$$B_\ell := \max_{N_\ell \leq j \leq N_\ell + m} d_j.$$

We first show that $B_{\ell+1} \leq B_\ell$ for all ℓ . Fix ℓ , and prove by induction on $s = 1, \dots, m+1$ that

$$d_{N_{\ell+1}+s-1} \leq B_\ell.$$

Set $k := N_\ell + m + s - 1$, so $k+1 = N_{\ell+1} + s - 1$. Any index used on the right-hand side of (1) lies in $[k - m_k, k]$, hence in

$$[N_\ell, N_\ell + m] \cup [N_{\ell+1}, N_{\ell+1} + s - 2].$$

By the induction hypothesis, all indices in the second set already satisfy $d_j \leq B_\ell$, and by definition all indices in the first set do too. Therefore the convex combination in (1) is bounded above by B_ℓ , and since the second term in (1) is nonpositive, we get

$$d_{N_{\ell+1}+s-1} \leq B_\ell.$$

This proves $B_{\ell+1} \leq B_\ell$. Hence $\{B_\ell\}$ is nonincreasing and bounded below by 0, so it is bounded. In particular, $\{x_k\}$ is bounded.

Next we prove that arbitrarily small residuals occur infinitely often. Suppose not. Then there exist $\varepsilon > 0$ and K such that

$$\|r(x_k)\| \geq \varepsilon \quad \text{for all } k \geq K.$$

Choose ℓ_0 such that $N_{\ell_0} \geq K$. Fix $\ell \geq \ell_0$. We again induct on $s = 1, \dots, m+1$, but now use the lower bound on the current weight: for $k = N_\ell + m + s - 1$, equation (1) gives

$$d_{k+1} \leq \sum_{i=0}^{m_k} \alpha_i^{(k)} d_{k-m_k+i} - \delta \alpha_{m_k}^{(k)} \|r(x_k)\|^2.$$

All d -terms in the convex combination are $\leq B_\ell$ by the same block-induction argument as above. Also, $\alpha_{m_k}^{(k)} \geq \underline{\alpha}$ and $\|r(x_k)\| \geq \varepsilon$. Hence

$$d_{k+1} \leq B_\ell - \delta \underline{\alpha} \varepsilon^2.$$

Therefore every point in block $\ell + 1$ satisfies

$$d_j \leq B_\ell - \delta \underline{\alpha} \varepsilon^2,$$

so

$$B_{\ell+1} \leq B_\ell - \delta \underline{\alpha} \varepsilon^2.$$

Iterating over $\ell \geq \ell_0$ would force $B_\ell \rightarrow -\infty$, impossible because $B_\ell \geq 0$. This contradiction proves: for every $\varepsilon > 0$, infinitely many k satisfy $\|r(x_k)\| < \varepsilon$.

Choose indices $k_j \rightarrow \infty$ such that $\|r(x_{k_j})\| \rightarrow 0$. Since $\{x_{k_j}\}$ is bounded in $\mathbb{R}^n$, it has a convergent subsequence, still denoted x_{k_j} , with

$$x_{k_j} \rightarrow \bar{x}.$$

Because g is nonexpansive, it is continuous, and therefore $r = g - I$ is continuous. Passing to the limit,

$$r(\bar{x}) = \lim_{j \rightarrow \infty} r(x_{k_j}) = 0.$$

Thus $g(\bar{x}) = \bar{x}$, so $\bar{x}$ is a fixed point. This proves part **(C)** and completes the theorem. $\square$

Gap to the full statement

What remains unproved here is a global theorem for the *exact, unconstrained* least-squares Anderson scheme on a broad nonlinear class *without* extra coefficient control.

Precisely:

- In part **(A)** I use the affine identity

$$\bar{r}_k = r(\bar{x}_k),$$

which is false for a general nonlinear map.

- In part **(B)** I obtain a rigorous global theorem for nonlinear contractions, but only under the explicit side condition

$$\sup_k q_k \|\alpha^{(k)}\|_1 < 1.$$

This condition is directly checkable from the computed least-squares coefficients, but I do *not* prove that the raw unconstrained least-squares solve automatically enforces it.

- In part **(C)** I obtain subsequential convergence for nonexpansive maps, but only under convex-combination and current-weight assumptions. Again, these assumptions are verifiable and can be enforced in constrained/safeguarded variants, but they are not automatic for the raw least-squares problem.
- Even under convex-combination assumptions in the nonexpansive case, I do not prove whole-sequence convergence for $m > 0$; only existence of a fixed-point cluster subsequence is proved here.

So the exact missing step to a full general theory is an a priori mechanism showing that the unconstrained least-squares coefficients of standard Anderson mixing remain globally controlled (or behave convexly enough) from properties of g alone.

Self red-team

- **Counterexample attempts against part (A).** I tried to break the affine proof by allowing M to have an eigenvalue 1. The proof itself rejects this: if $Mv = v \neq 0$, then $((1 - \beta_k)I + \beta_k M)v = v$, so the hypothesis $\sup_k \|(1 - \beta_k)I + \beta_k M\| < 1$ cannot hold. Thus the hidden invertibility issue for $I - M$ is genuinely resolved.
- **Counterexample attempts against part (B).** I tried to remove the factor $\|\alpha^{(k)}\|_1$. The proof breaks immediately: after rewriting in α -form, the triangle inequality produces exactly that factor, and there is no cancellation mechanism available in a general norm estimate. So the coefficient-control hypothesis is not cosmetic; it is the sharp obstruction in this line of argument.
- **Counterexample attempts against part (C).** I tried to drop the lower bound $\alpha_{m_k}^{(k)} \geq \underline{\alpha}$. The descent argument then no longer forces the current residual $\|r(x_k)\|$ to appear with positive weight in the right-hand side of (1), so the contradiction argument fails. This objection is valid, and is why I kept the current-weight hypothesis explicit.
- **Three likely referee objections.**
 1. *“The Γ -to- α conversion may have an indexing bug.”* I addressed this by writing out the telescoping identities in full in Lemma 1.
 2. *“The block recursion might accidentally depend on points older than one block.”* This is why Lemma 3 uses the shifted index $k = N_\ell + m + s - 1$; the interval inclusion is checked explicitly.
 3. *“The nonexpansive argument may only prove boundedness, not proximity to a fixed point.”* Correct: boundedness alone is not enough, so I proved the stronger statement that arbitrarily small residuals occur infinitely often, and only then extracted a convergent fixed-point subsequence.
- **Boundary and degenerate cases checked.**
 1. $m = 0$: part (B) reduces to relaxed Picard iteration for contractions; part (C) reduces to the familiar averaged-iteration inequality.
 2. $c = 0$: the contraction theorem predicts immediate convergence when $\beta_k \equiv 1$, which is correct because then g is constant.
 3. $g = I$: residuals vanish identically. In the nonexpansive regime this shows why whole-sequence convergence cannot be expected without stronger selection rules when the fixed-point set is large and the least-squares problem is degenerate.
 4. $\beta = 1$: in the contractive setting this is the best constant damping in my estimates, since $q(\beta)$ is minimized at $\beta = 1$.
- **Did I accidentally prove a false stronger claim?** No. The proof of part (B) does *not* show that raw Anderson mixing always converges for contractions; it only shows convergence when the effective gain $q_k \|\alpha^{(k)}\|_1$ stays uniformly below 1. This avoids conflict with the literature’s repeated warnings about coefficient ill-conditioning in unrestricted AA.

Ideas tried

1. Direct residual-monotonicity for general nonlinear maps.

Idea. Since $\Gamma = 0$ is feasible, the least-squares step gives $\|\bar{r}_k\| \leq \|r(x_k)\|$. I first tried to turn this into a direct residual contraction.

Obstruction. For nonlinear g , $\bar{r}_k$ is generally *not* equal to $r(\bar{x}_k)$. The exact affine identity

$$\bar{r}_k = r(\bar{x}_k)$$

is what makes part **(A)** work, and it fails outside the affine case.

Next step. Find a nonlinear substitute for this identity (for example via higher-order residual expansions or a safeguard ensuring approximate consistency). That is exactly the direction taken in some one-step analyses and safeguarded variants.

2. Operator-theoretic / Fejér analysis under nonexpansiveness without coefficient restrictions.

Idea. Rewrite the Anderson step as a combination of averaged-operator images and try to prove quasi-Fejér monotonicity.

Obstruction. Negative coefficients destroy convexity, so

$$\left\| \sum_i \alpha_i y_i \right\|^2 \leq \sum_i \alpha_i \|y_i\|^2$$

is unavailable. This blocks the standard averaged-operator descent argument.

Next step. Either constrain the coefficients (which leads to part **(C)**), or derive a different Lyapunov functional that can tolerate sign-indefinite coefficients.

3. Scalar depth-one reduction to the secant method, in search of a disproof.

Idea. For $n = 1$ and $m = 1$, whenever $r(x_k) \neq r(x_{k-1})$, the least-squares problem is one-dimensional and forces

$$\bar{r}_k = 0,$$

so the Anderson update becomes exactly

$$x_{k+1} = x_k - r(x_k) \frac{x_k - x_{k-1}}{r(x_k) - r(x_{k-1})},$$

the classical secant method applied to $r(x) = 0$.

Obstruction. This reduction makes a global counterexample plausible, but constructing a *contractive fixed-point map* g for which the induced secant dynamics fails globally turned out subtler than I could settle rigorously here.

Next step. Either produce an explicit scalar contraction counterexample via secant dynamics, or prove a scalar global theorem for strongly monotone residuals. Either direction would materially sharpen the present partial result.

4. Importing GMRES/Arnoldi linear theory.

Idea. Use Walker–Ni’s linear equivalence to GMRES/Arnoldi as the main proof of the affine case.

Obstruction. That theory is cleanest in full memory, while the prompt asks for finite memory m . Rather than import a proof with mismatched hypotheses, I rederived part **(A)** directly from residual affinity.

Next step. A refined spectral analysis of the finite-memory affine case might produce sharper rates than my simple ρ -bound.

Invoked results

No external research theorem is used as a step inside the formal proof.

All proof steps are derived in-line. The only ambient finite-dimensional facts used are:

- bounded sequences in $\mathbb{R}^n$ have convergent subsequences;
- the squared Euclidean norm is convex.

I therefore have no citation-hypothesis audit to report for the proof body itself. All web-cited papers above were used only for orientation, classification of prior results, and literature hygiene.

Self-confidence

3 — I checked every proof step carefully and red-teamed the main inferences; the affine and contractive parts are especially solid, while the nonexpansive subsequence argument is more delicate but I could not find a hole in it.

Bibliography

```
\begin{verbatim}
@article{Anderson1965,
  author = {Anderson, Donald G.},
  title = {Iterative Procedures for Nonlinear Integral Equations},
  journal = {Journal of the ACM},
  volume = {12},
  number = {4},
  pages = {547--560},
  year = {1965},
  doi = {10.1145/321296.321305}
}

@article{WalkerNi2011,
  author = {Walker, Homer F. and Ni, Peng},
  title = {Anderson Acceleration for Fixed-Point Iterations},
  journal = {SIAM Journal on Numerical Analysis},
  volume = {49},
  number = {4},
  pages = {1715--1735},
  year = {2011},
  doi = {10.1137/10078356X}
}

@article{TothKelley2015,
  author = {Toth, Alex and Kelley, C. T.},
  title = {Convergence Analysis for Anderson Acceleration},
```

```

journal = {SIAM Journal on Numerical Analysis},
volume   = {53},
number   = {2},
pages    = {805--819},
year     = {2015},
doi      = {10.1137/130919398}
}

@article{ZhangODonoghueBoyd2020,
author   = {Zhang, Junzi and O'Donoghue, Brendan and Boyd, Stephen},
title    = {Globally Convergent Type-I Anderson Acceleration for Nonsmooth Fixed-Point Iterations},
journal  = {SIAM Journal on Optimization},
volume   = {30},
number   = {4},
pages    = {3170--3197},
year     = {2020},
doi      = {10.1137/18M1232772}
}

@article{PollockRebholz2021,
author   = {Pollock, Sara and Rebholz, Leo G.},
title    = {Anderson acceleration for contractive and noncontractive operators},
journal  = {IMA Journal of Numerical Analysis},
volume   = {41},
number   = {4},
pages    = {2841--2872},
year     = {2021},
doi      = {10.1093/imanum/draa095}
}

@article{BianChenKelley2021,
author   = {Bian, Wei and Chen, Xiaojun and Kelley, C. T.},
title    = {Anderson Acceleration for a Class of Nonsmooth Fixed-Point Problems},
journal  = {SIAM Journal on Scientific Computing},
volume   = {43},
number   = {5},
pages    = {S1--S20},
year     = {2021},
doi      = {10.1137/20M132938X}
}

@article{BarreTaylordAspremont2022,
author   = {Barr{\`e}, Mathieu and Taylor, Adrien and d'Aspremont, Alexandre},
title    = {Convergence of a Constrained Vector Extrapolation Scheme},
journal  = {SIAM Journal on Mathematics of Data Science},
volume   = {4},
number   = {3},
pages    = {979--1002},
year     = {2022},
doi      = {10.1137/21M1428030}
}

@article{Saad2025,
author   = {Saad, Yousef},
title    = {Acceleration methods for fixed-point iterations},
journal  = {Acta Numerica},
volume   = {34},
pages    = {805--890},

```

```
year    = {2025},  
doi     = {10.1017/S0962492924000096}  
}  
\end{verbatim}
```